Earth and Planetary Science Letters, 106 (1991) 133–150
Elsevier Science Publishers B.V., Amsterdam

[XLeP]

Biostratigraphy and magnetostratigraphy of the Cretaceous/Tertiary Sopelana section (Basque country)

Claire Mary [a], Marie-Gabrielle Moreau [a], Xavier Orue-Etxebarria [b], Estibaliz Apellaniz [b] and Vincent Courtillot [a]

[a] *Laboratoire de Paléomagnétisme et Géodynamique, Département de Géomagnétisme (UA 729 CNRS), Institut de Physique du Globe de Paris, Paris, France*

[b] *Dpto. de Geologia, Fac. de Ciencias, Universidad del Pais Vasco, Bilbao (Pais Vasco), Spain*

Received December 21, 1990; revised and accepted June 10, 1991

ABSTRACT

Remarkably thick sequences of upper Cretaceous and lower Tertiary sediments outcrop on the northern coast of the Basque country. We have sampled the well exposed section along the Sopelana beach over 150 m, spanning roughly from middle Maastrichtian (71 Ma) to lower Paleocene (66 Ma), for both biostratigraphic and magnetostratigraphic studies.

Thermal demagnetization of ~400 specimens revealed both normal and reversed recent overprints unblocked below 200°C, and both normal and reversed characteristic directions at higher temperatures (200–450°C in the grey limestones; 350–550°C in the redder marls and limestones). Thermal and AF demagnetization and monitoring of weak field susceptibility are consistent with some form of (titano-) magnetite as the main carrier of magnetization. 259 demagnetization diagrams yielded two nearly antipodal clusters of directions, which are still polluted by some 20% remaining recent overprint. The overall mean direction in stratigraphic coordinates is $D = 356°$, $I = 51°$ ($\alpha = 3°$), consistent with what is expected for Eurasia at KTB time. Magnetic stratigraphy outlines a succession of eight polarity intervals and is rather straightforward, except for two highly complex zones (52 to 40 m, and 28 to 26 m below the KTB) where the magnetic polarity appears to flip at an unreasonable rate. Biostratigraphic check allows unambiguous assignment of several chrons, with the recognition of the *Gansserina gansseri, Abathomphalus mayaroensis, "Globigerina" eugubina, Eoglobigerina edita* (= *E. pseudobulloides*) and *E. trinidadensis* zones. The magnetostratigraphic section therefore begins in chron 31R and ends in 29N. Further detailed study of the complex zones reveals that specimens there have a distinct magnetic behaviour: susceptibility increases beyond 400°C, indicating instability and mineralogical change, intensities are higher than elsewhere and more scattered, and a higher unblocking temperature/higher coercivity magnetic component is uncovered. This component, which is absent from the rest of the section, could be an early chemical remagnetization. Indeed the complex zones contain larger amounts of chlorite, a sign of more intense diagenesis. We have therefore discarded samples showing this anomalous behaviour, and have retained only the 223 samples where demagnetization was achieved by 450°C. The thicker, lower complex zone reduces to a reversed chron which we propose to correlate with 30R. We have identified an extra short reversed event within chron 30N, for which independent support is found in DSDP Sites 524 and 577a, and possibly in Gubbio. Finally, the high-sedimentation rate section at Sopelana (25 m/Ma in the Maastrichtian, 7 m/Ma in the Danian) displays one of the best resolved on-land magnetostratigraphies around KTB time. The two complex zones of early remagnetization form an "echo" which may have been generated some 100 kyr after deposition of the sediments. The KTB itself occurs approximately halfway to 3/5 up 29R in terms of time.

1. Introduction

The biological events that took place at the Cretaceous–Tertiary boundary have attracted much attention to stratigraphic sections that have recorded the events in as much detail and with as much fidelity as possible. A quite fine chronology of geochemical and biostratigraphical events is now available. However, the detailed behaviour of the geomagnetic field, potentially the most accurate tool for calibration and determination of the simultaneity of events at the global scale, is not yet known to the degree of detail that might be desirable. Although it has been established for over a decade that the Cretaceous–Tertiary boundary (KTB) occurs within the magnetic chron 29R [1, 2], large and systematic variations in sedimentation rate do not allow yet a precise absolute

0012-821X/91/$03.50 © 1991 – Elsevier Science Publishers B.V. All rights reserved

positioning of the KTB within 29R and accordingly an independent correlation of biological and geochemical anomalies within this zone. In addition, some magnetostratigraphic studies of DSDP cores reveal elusive and intriguing short magnetic events within well established magnetic chrons [3–5]: if confirmed, these would provide precious additional markers for correlation and also information on the detailed behaviour of the geomagnetic field at that time.

The remarkably thick sequences which outcrop on the northern coast of the Basque country, both in Spain and France, have long drawn attention and have been the focus of biostratigraphic, geochemical and magnetostratigraphic studies [6–14] (see Fig. 1 for locations). Magnetostratigraphic results have been obtained for lower Paleocene sections in Zumaya and San Sebastian [10]. The KTB part of the Bidard section [14] is difficult to interpret, due either to a gap in sedimentation or to a major remagnetization of the uppermost Maastrichtian and lowermost Danian strata. The Zumaya section has not yielded interpretable paleomagnetic results in its Maastrichtian part (K. Verosub, unpublished work and pers. commun., 1986). We have focused on the well exposed section along the beach at Sopelana (Fig. 1) in the hope of completing its biostratigraphy [15, 16] and obtaining a reliable magnetostratigraphy from the middle Maastrichtian into the lower Paleocene.

2. Geological setting

The large Bilbao synclinorium (Fig. 1) belongs to the Basco–Cantabric basin which borders the Bay of Biscay to the west of the Pyrenees. It consists of thick sequences of alternating marls and limestones spanning from the Albian (Cretaceous) to the middle Eocene (Tertiary). The marls and limestones (and some turbidites) were folded in the Eocene (about 40 to 50 Ma) during a major tectonic phase which resulted from the collision of Africa with Europe [e.g. 17, 18]. Although fully continuous sections are not available around the syncline, there are many outcrops which are apparently continuous over several geological stages. In particular, the KTB has been located precisely at several sites along the coast. In Sopelana, the upper Cretaceous and lower Paleocene crop out as a long, 10 m high cliff, not far from Bilbao [18]. The section has already yielded abundant inoceramids [19], planktonic foraminifera and coccoliths. The bio-zonation of the Paleocene foraminifera and nannoplankton has been de-

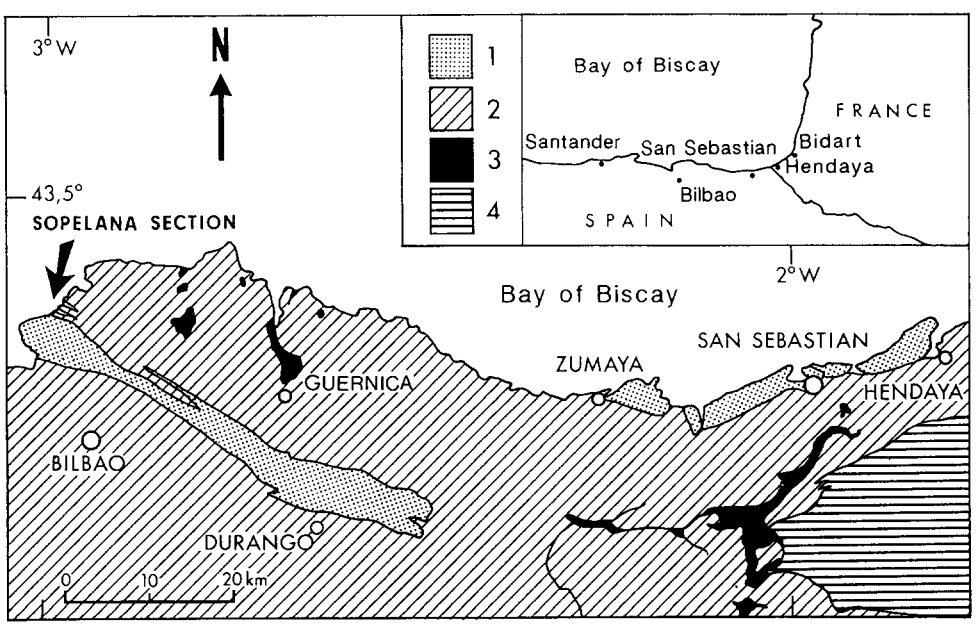


Fig. 1. Schematic map of Bay of Biscay sections from Plaziat [17] and Rat [18].

termined in 1983 (Fig. 2, Table 2; [15, 16]). Prior to the present work, the Maastrichtian part was not so well known, with only the *A. mayaroensis* Zone being identified from a few samples. Inoceramids vanish just below this zone, whereas ammonites (which are rather few) survive until the KTB [19].

The outcrop appears to dip rather regularly by 70° to the SSW in monoclinal fashion, although plunge of fold axes at a larger scale remains a possibility. There was no obvious large-scale slumping or fault. The only clear offset, less than 3 m, was found 95 m below the KTB. The Maastrichtian part of the section consists of grey to pink marly limestones that become pinker as the KTB is approached. Thick sequences of alternating limestone and marl beds of similar thickness (about 30 cm) are interrupted by three redder

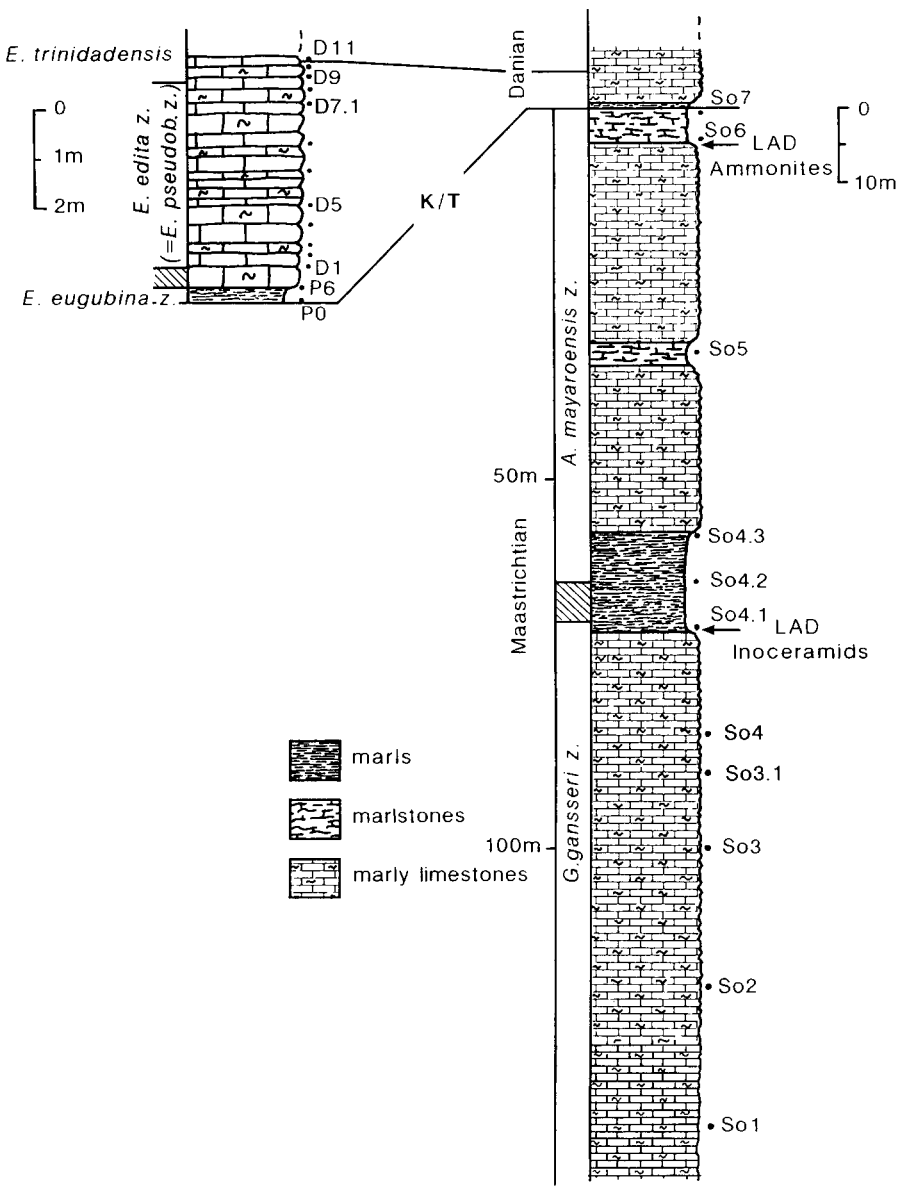
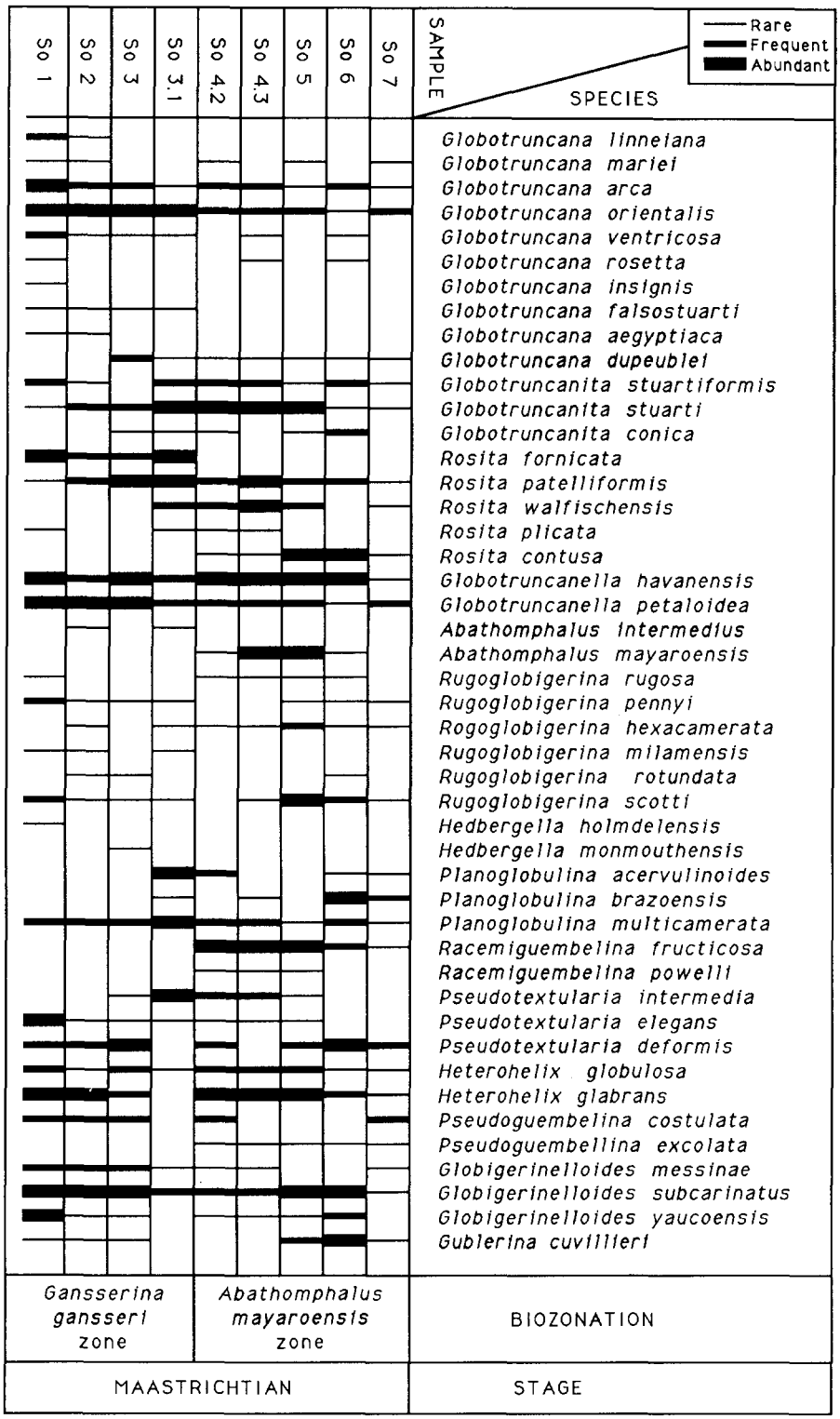


Fig. 2. Lithofacies and biostratigraphy of the Sopelana section, from Lamolda et al. [15], Ward [19], and this paper. Palaeontological samples are given with their code names (from So1 to D11).

TABLE 1

Planktonic foraminifera observed in the Maastrichtian part of the Sopelana section



marly zones (Fig. 2). *A. mayaroensis* appears in the lower one, which is 15 m thick; the upper one is 5 m thick and ends below the KTB.

The boundary is marked by a clay level (a few cm thick) followed by 30 cm of green marl. In the Danian part of the section, the limestone beds become thicker and pinker and interbedded marls are thin or practically non-existent. Based on biostratigraphy (Fig. 2), the sedimentation rate was thought to have dropped from 30 m/Ma in the Maastrichtian to 4 m/Ma in the Danian. These rate measurements [15] were based on the thickness of the *A. mayaroensis* and *E. trinidadensis* zones and the mean sedimentation rates estimated for the Gubbio section. If one follows the Kent and Gradstein time-scale [20, 21], the sedimentation rates at Sopelana become 25 m/Ma for the Maastrichtian and 7 m/Ma for the Danian, still a drop by a factor of 3.5 across the KTB.

3. Biostratigraphy

The samples used for the biostratigraphic study of Maastrichtian planktonic foraminifera from the Sopelana section have been taken in marlstones and marly limestones that appear in the stratigraphic series exposed along the beach: these are generally grey or sometimes red. Most are rich in planktonic foraminifera, the poorest being the uppermost one (So7, Fig. 2; Table 1). The proportion of planktonic foraminifera with respect to the total foraminifera in the different samples is greater than or equal to 80%. The size of fractions studied lies between 0.1–0.5 mm and 0.5–1 mm, except for sample So7, in which the fraction with size smaller than 0.1 mm was analyzed and no form similar to *Guembelitria cretacea* was found.

3.1. Maastrichtian

Two Maastrichtian zones (Fig. 2, Table 1) are identified: the *Gansserina gansseri* zone and the *Abathomphalus mayaroensis* zone. The first zone includes the first five samples (So1 to So4.1), until the appearance of *A. mayaroensis* in sample So4.2, which corresponds to the next zone. Although the index species *G. gansseri* was not found in any of the samples studied, the assemblages present in the first five samples belong without any doubt to the *G. gansseri* zone. The remaining samples (So4 to So7) belong to the *A. mayaroensis* zone, although it could have been included in the informal *G. stuarti* zone [22] or *Pseudotextularia deformis* zone [23], since the last sample (So7) does not contain its index species.

The most commonly observed species within the Maastrichtian series at Sopelana are *Globotruncana arca, G. orientalis, Globotruncanita stuarti, Rosita patelliformis, Globotruncanella havanensis, G. petaloidea, Planoglobulina multicamerata, Heterohelix globulosa, H. glabrans* and *Globigerinelloides subcarinatus*. On the other hand, typical species such as *R. contusa* and *Racemiguembelina fructicosa* are only found at the beginning of the *A. mayaroensis* zone, along with the index species. The final sample So7, recovered 2 cm below the KTB, shows a large number of species, and contains an assemblage characterized by *G. arca, G. orientalis, G. stuarti, R. patelliformis, R. contusa, G. havanensis, G. petaloidea, R. scotti, R. fructicosa, P. deformis* and *Gublerina cuvillieri*.

3.2. Paleocene

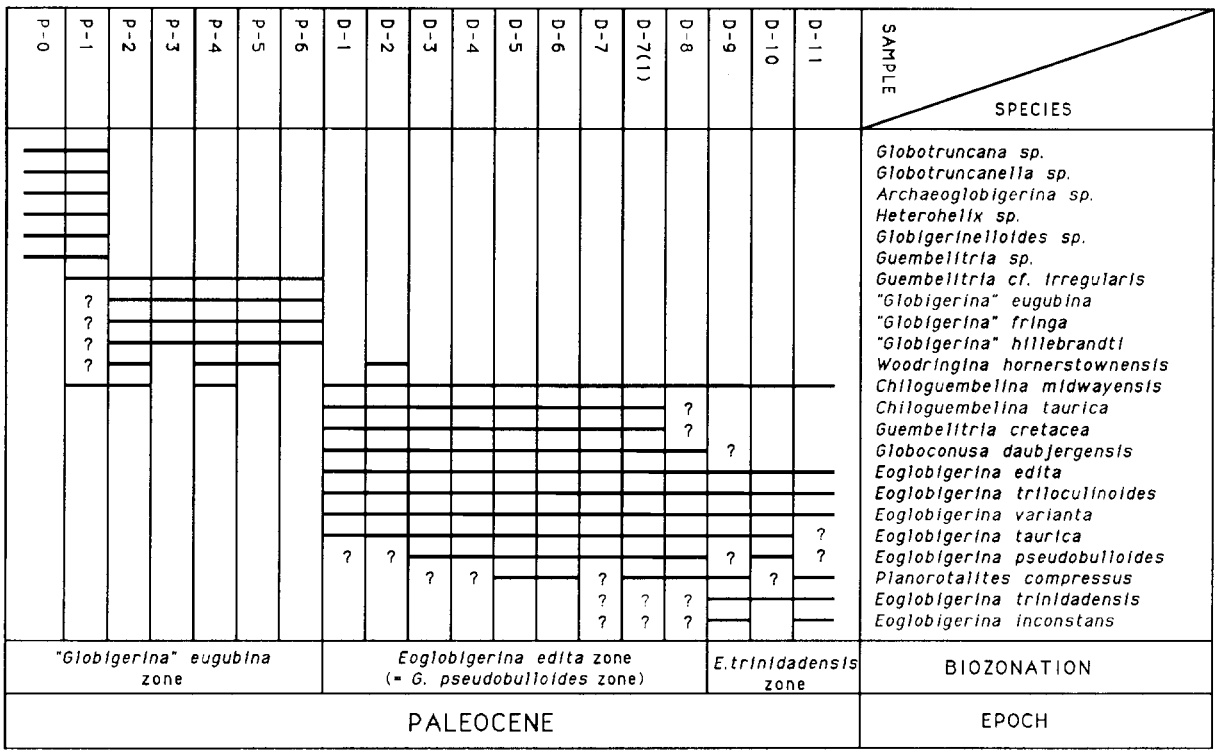
The samples from the base of the Paleocene part of the section are poor in planktonic foraminifera, as is observed in samples of the same age from other sections in the Basco–Cantabric Basin (Fig. 2, Table 2). Furthermore, specimen preservation is in most cases rather poor and the proportion of planktonic foraminifera with respect to total foraminifera content of the sample varies greatly. The proportion reaches "normal" values (i.e. above 75%) in the middle part of the *E. trinidadensis* zone.

Three zones are identified; the first one, the "*Globigerina*" *eugubina* zone, comprises the boundary clay-level and the first 30 cm of grey marlstones up to the Danian marly limestones (Fig. 2). Neither the clay (sample P-0), nor P-1 contain characteristic planktonic foraminifera of the Tertiary, with the exception of a certain form included within the genus *Guembelitria*. The remaining samples of this zone, from P-2 to P-6, show a typical assemblage with "*G.*" *eugubina*, "*E.*" *fringa*, "*E.*" *hillebrandti* and *Guembelitria* cf. *irregularis* among others.

Above this zone, a series of (generally pink) alternating limestones and marly limestones marks the beginning of the *Eoglobigerina edita* zone,

TABLE 2

Planktonic foraminifera observed in the Danian part of the Sopelana section



with the appearance of species such as *E. trilobulinoides, E. varianta, E. taurica, Globoconusa daubjergensis* and *Guembelitria cretacea*, in addition to the index species. *Planorotalites compressa* is found from approximately the middle part of this zone upwards. Beginning with sample D-9, typical forms of *E. trinidadensis*, marking the beginning of the zone bearing that name, are found along with individuals of *E. inconstans*. The rest of the assemblage is very similar to that in the previous zone.

4. Paleomagnetism

4.1. Sampling

In a first session of fieldwork, only well consolidated limestone beds were sampled, every two meters (corresponding roughly to 75 ± 10 kyr) in the Maastrichtian and every single bed in the Danian. The sampled section extended from 140 m below the KTB to 5 m above. Because of unfortunate gaps in the data and also of previously unsuspected difficulties revealed by the magnetic analysis, a second session of fieldwork was carried out: further sampling was performed, for instance in a landslide area, 95 m below the KTB, where beds were sampled at the top rather than at the bottom of the cliff after checking for lateral continuity, and in the fragile marlstones below the KTB, where oriented blocks were sampled for further drilling in the laboratory. Most of the samples were however collected with an electric drill and oriented with a magnetic compass.

4.2. Measurements

The 2.5 cm long cores were cut to a length of 2.2 cm and their magnetization was measured with a CTF squid magnetometer. Demagnetization was

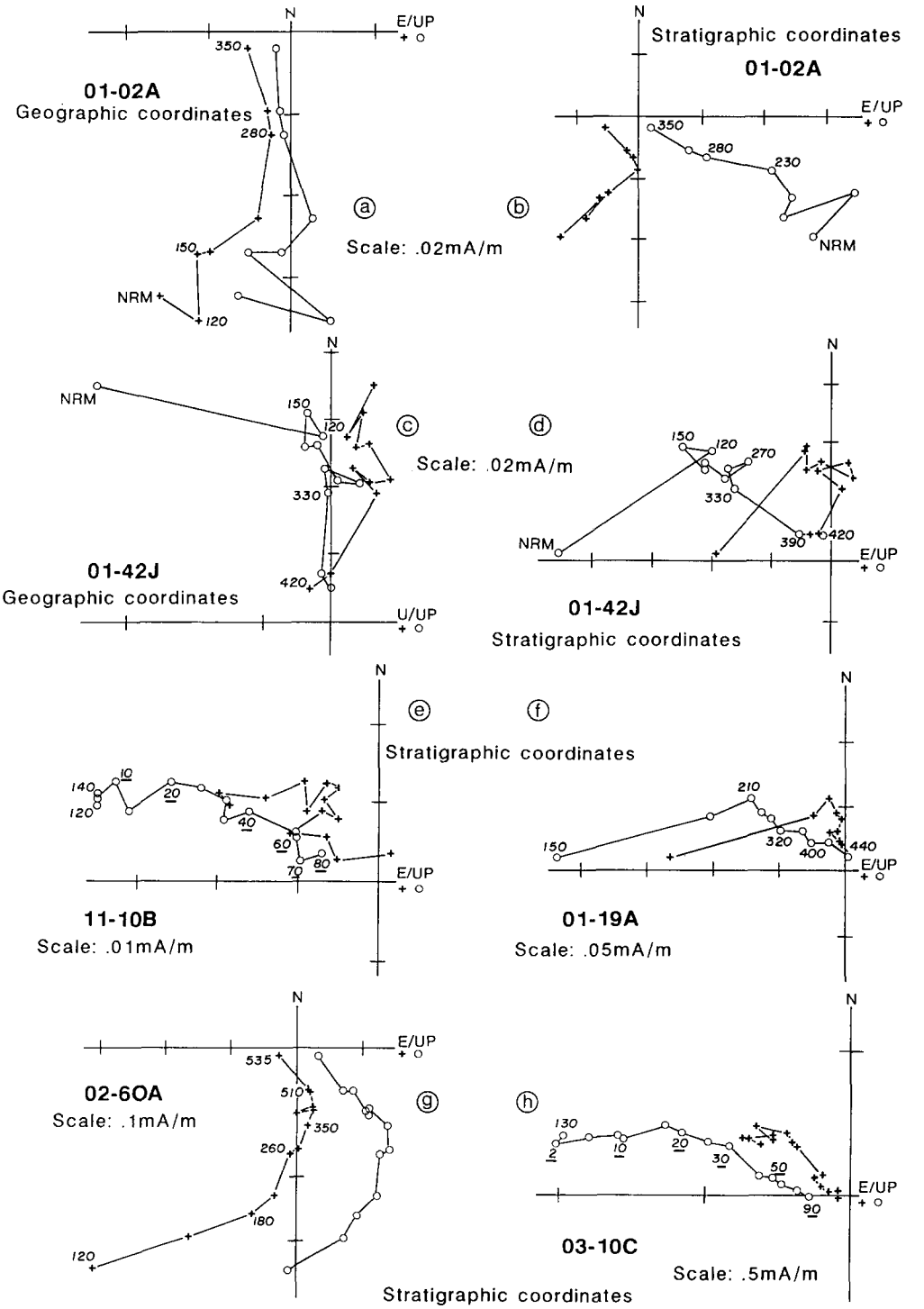


Fig. 3. Representative vector demagnetization diagrams from specimens of the Sopelana section. The specimen identification is given near the abscissa axis. Crosses are for the horizontal projection and open circles for the projection in the NS vertical plane. (a), (b), (c), (d), (e), (f): grey Maastrichtian marlstones; (g): red marlstones beneath the KTB; (h): reddish Danian marlstones. (e) and (h) are alternating field demagnetizations (values in mT), after thermal demagnetization up to 140°C; the others are thermal demagnetizations (values in °C). (a) and (c) are displayed in in-situ (geographic) coordinates, and (b), (d), (e), (f), (g), and (h) after tectonic correction (stratigraphic coordinates).

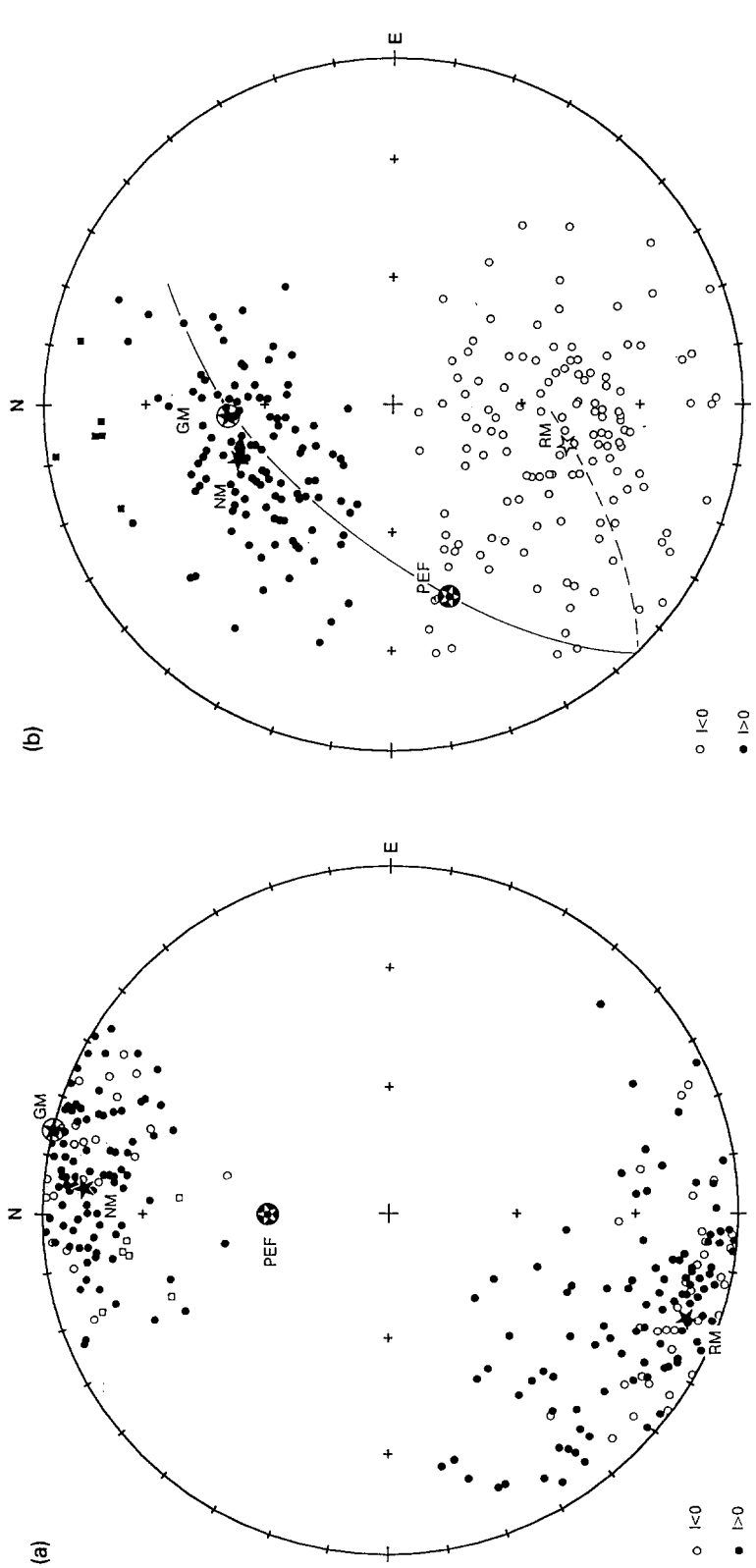


Fig. 4. Lower hemisphere equal area projection of characteristic directions before (a) and after (b) tectonic correction. Open symbols, reversed directions; closed symbols, normal directions (squares are intermediate directions from sampling level 66). *PEF*: present earth's dipole field. *GM*: global mean. *NM*: normal mean. *RM*: reversed mean.

most often achieved in a laboratory-built zero field ($\leqslant 10$ nT) furnace in steps up to 450°C beyond which samples were either demagnetized or became altered and remagnetized. AF demagnetization, performed with a Schönstedt apparatus, generally gave results that were much noisier but consistent with their thermal counterparts. Some samples, which were not demagnetized under 550°C, were very resistant to AF cleaning. Magnetic changes upon heating were monitored by measuring the weak-field susceptibility of the samples. Natural remanent magnetization prior to treatment was on the order of 2 to 8 mA/m in the marls below the KTB and in the Danian limestones, and about 0.1 mA/m in the Maastrichtian limestones.

4.3. Paleomagnetic directions

Figure 3 gives some examples of demagnetization diagrams. Although many similar diagrams were obtained out of the 400 specimens which were measured, they are a selection rather than a random sampling, since there was a fair amount (about 10%) of somewhat noisier diagrams. In most cases, a first component of magnetization roughly in the direction of the present earth field (PEF) was removed by 120°C in the Maastrichtian limestones (e.g. Fig. 3c) and 200°C in the marlstones and Danian marls (e.g. Fig. 3, f and g). In many cases, a second component opposite to the PEF was removed between 120 and 150°C (e.g. Fig. 3, a and c). These two components are recent overprints and were acquired later than the folding of the series. In the grey Maastrichtian limestones, a characteristic component decays roughly to the origin: the last demagnetization steps correspond to weak intensities and yield only approximate directions (Fig. 3, a and e). This component was uncovered between 200–300°C and 350–450°C (e.g. Fig. 3b; Fig. 3d, where little magnetization was removed between 150 and 300°C and decay occurred between 330 and 420°C; Fig. 3f, with a single characteristic component removed from 210 to 440°C). A similar characteristic direction was found between 350 and 550°C in red marls below the KTB (Fig. 3g) and in the redder Danian limestones. Most of samples could be demagnetized between 30 and 80 mT (Fig. 3, e and h), where they display the same component. The characteristics of thermal and AF demagnetization apparently allow us to eliminate hematite or sulfides as potential magnetic carriers of the characteristic direction (except in the Danian samples, in which hematite appears to carry a PEF component which remains after heating up to 550°C). The fact that weak field susceptibility remains constant through about 350°C (outside of the "anomalous zones") argues against sulfides or maghemite. It therefore seems that the dominant magnetic carriers might be some form of magnetite or titanomagnetite with various grain sizes.

Whenever possible, the characteristic direction was determined by principal component analysis [24], with linear segments fit by least squares to at least 4 points (often more). In some cases, the magnetization remained locked over a wide temperature range, and the intensity fell to zero close to the Curie temperature of magnetite. In that case, a Fisher average of the directions in the "locked range" was determined. All these peculiar samples come from two well determined zones (respectively 23 and 41 m below the KTB, and 2 and 12 m thick) to which we will return later.

About 150 samples were discarded, either because the PEF component completely masked the primary component, or because several distinct components were present and we felt unable to unequivocally identify a single characteristic (primary?) direction, or because of peculiar behaviour discussed below within complex zones A and B. The remaining directions form two roughly antipodal clusters with rather large scatter but comparatively few intermediate directions. The directions are remote from the PEF (Fig. 4a, geographic coordinates). When rotated to stratigraphic coordinates (Fig. 4b), the clouds are found to be somewhat elongated along a great circle going through the PEF. The means of the normal and reversed directions are not antipodal at the 99% confidence level, but are both significantly displaced on the same great circle, in the direction of the PEF. This indicates that the characteristic directions obtained through demagnetization still contain some amount of PEF overprint due to overlapping spectra of the grains carrying the different components. Because the numbers of normal and reversed samples are essentially the same (125 and 134, respectively), the bias due to the

overprint will be cancelled in a global average. The overall mean direction is $D = 356°$, $I = 51°$ ($K = 10$, $\alpha = 3°$), which is fully consistent with the direction expected for that time and location from a synthetic apparent polar wander path of Eurasia [25]: $D = 358 \pm 5°$, $I = 51 \pm 5°$. Using the angular deviation between the PEF and both normal and reverse means, one can show that the amount of PEF overprint that contaminates the original magnetization is on the order of 20% of the intensity of the primary component. In addition to providing a new paleomagnetic datum for the Eurasian APWP, this first phase of the study seems to imply that a reliable sequence of normal and reversed primary magnetic directions has been uncovered in the Sopelana KTB section.

4.4. *Magnetostratigraphy*

The characteristic directions shown in Fig. 4 have been plotted (in stratigraphic coordinates) as a function of the sampling level with respect to the

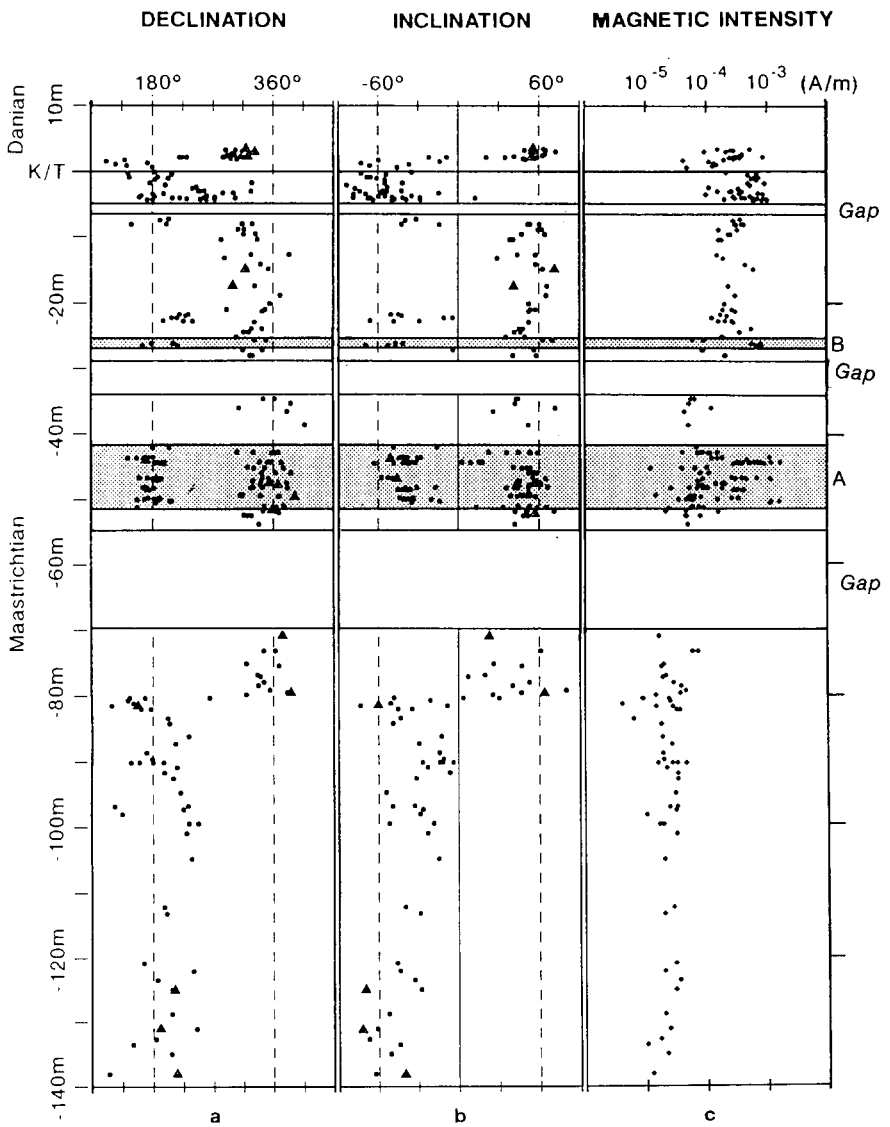


Fig. 5. (a) Declination and (b) inclination of characteristic magnetic directions. Closed circles are for directions resulting from thermal demagnetization, triangle symbols for directions from alternative field demagnetization. (c) Magnetic intensity remaining in each specimen after 300°C cleaning. Shaded portions: complex zones A and B. White portions: sampling gaps.

KTB in Fig. 5, together with the remaining NRM after heating at 300°C. Most directions were indeed obtained by thermal treatment (dots), although a few samples were AF demagnetized (triangles): the two kinds of data are seen to be consistent at all levels (Fig. 5). A 60 m thick sequence of reversed polarity extends from the base of the section (−140 m) to about −80 m. A 10 m thick sequence of normal polarity follows. Despite some scatter, there are few intermediate points. Magnetic intensity after heating at 300°C is rather uniform at about 2.10^{-5} A/m. Because we are in the *G. gansseri* zone, with the *A. mayaroensis* zone beginning near −64 m, it is clear that the lower reversed zone is 31R and the normal one above the beginning of 31N [1]. Above this, unfortunately there is a ~15 m wide gap where very fragile marls could not be sampled (see Fig. 2). Above the gap, from −54 to −52 m, we find normal samples whose characteristics are very similar to those below the gap. From −52 to −40 m, we encounter a complex zone where magnetic polarity appears to flip frequently and magnetic intensity fluctuates, with values up to 1 mA/m. This 12 m thick complex zone will be called "A" and is studied further below. Another such complex zone is encountered between −28 and −26 m ("B"), and is also the subject of a separate discussion. Apart from these two zones, magnetic behaviour is rather straightforward, with a long normal zone from −40 m to −8 m, interrupted by a clear, 1 m thick reversed zone at −22 m. From 8 m below the KTB to 1.8 m above, magnetic polarity is reversed as expected, with a magnetic intensity after heating at 300°C between 0.1 and 1 mA/m. Some fluctuations occur above this, in a 50 cm range, until magnetic polarity clearly shifts to normal (and again stable) from 2.3 to 3.4 m above the KTB (Fig. 5). Above this, demagnetization diagrams become extremely noisy, with evidence for at least four components of magnetization, opposite in direction by pairs with overlapping unblocking temperature spectra. We were not able to extract convincing and reliable primary directions from these samples. We note a few samples below the KTB (−2 to −5 m) which have a reversed inclination and an awkward declination. We suspect that, due to plasticity of the marls, some bedding-parallel deformation may have taken place during folding. Bed over bed rotational slippage would mainly alter declinations and not inclinations. In addition, one isolated sample displays a positive inclination; despite renewed denser sampling, no other such sample was found and the observation could not be duplicated. Hence this "normal" polarity result appears to be suspect.

In summary, three magnetic chron boundaries are clearly determined in the Sopelana section: 31R/31N at −80 m, 30N/29R at −8 m and 29R/29N at +2 m. The KTB occurs 80% up chron 29R, which is 10 m thick (i.e. over twice as thick as in the Gubbio reference section). If we assume that the average sedimentation rates determined over a much larger thickness apply during chron 29R, we find that the Maastrichtian part lasted some 320 kyr and the Danian one 290 kyr for a total of some 600 kyr, which is consistent with the generally accepted value [20, 26]. The KTB which, in the most complete and continuous sections, occurs 2/3 to 3/4 up chron 29R in stratigraphic height, actually occurs halfway to 3/5 up in terms of time. A proposed chronometer based on the identification of "Milankovitch cycles" yields a boundary 3/5 up chron 29R in terms of time [27]. A significant uncertainty remains, considering both the differences between reference time scales [1, 20, 21, 26] and the possible changes in sedimentation rate within an individual chron, and particularly 29R.

In the Sopelana section, chron 30R is not unambiguously determined; it could either correspond to the clear reversed interval at −22 m, which appears to be very short (35 ± 5 kyr if the average sedimentation rate applies), or be hidden within complex zone A. We have pointed out in the introduction the potential interest of discovering short chrons within longer, well-established ones. Unfortunately, the complex behaviour found in several segments of what should be the 31N/30R/30N sequence cast doubt on the overall reliability of this magnetostratigraphy and clearly warranted further analysis.

4.5. Study of complex zones A and B

The complex behaviour observed in zone A after the first field-session prompted us to return to the site for a second session of much denser sampling. The result was an even more bewilder-

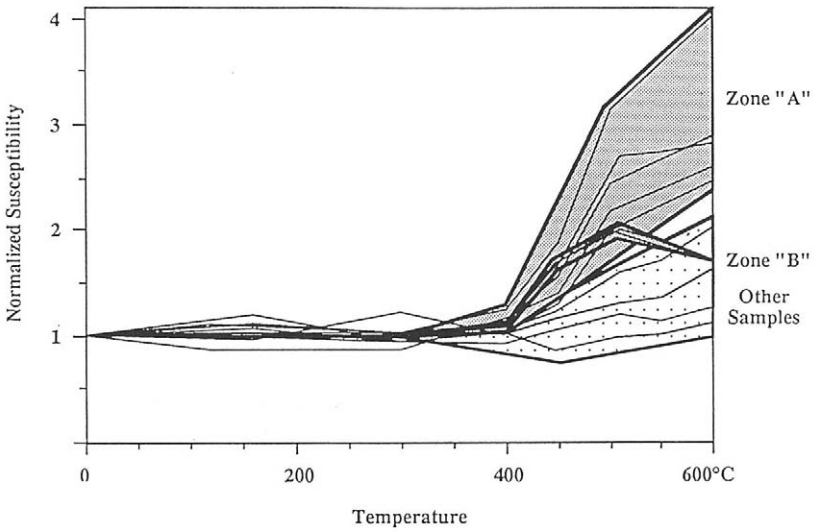


Fig. 6. Normalized susceptibility as a function of reheating temperature: shadings correspond to samples from complex zones A and B.

ing pattern seen between −52 and −42 m (Fig. 5). The observed pattern cannot have resulted from true field behaviour. Because of the magnetic polarity observed both above and below, and because of the available biostratigraphic correlation, the zone must be located within 30N or 31N, with a strong suggestion that 30R is "hidden" somewhere inside of it. The alternate pattern of normal and reversed polarity can result from two equally complex patterns of either normal remagnetization in 30N time of rocks deposited during 30R, or of reversed remagnetization during 30R of rocks deposited during 31N. Remagnetization could of course have taken place later, for instance during 29R. In case chron 30R should rather be correlated with the 1 m thick reversed interval at −22 m, the same hypothesis could still be formulated for zone A, assuming the occurrence of an as yet unknown reversed chron within 31N.

It is particularly striking that the characteristic directions are so nearly antipodal, and that no intermediate directions have been found, except for six samples, all located within a 10 cm interval at −43 m. These samples do yield well grouped intermediate directions (squares in Fig. 4, $\alpha_{95} = 10°$), which can hardly be interpreted as an actual transitional field. This represents only 1% of the suspect zone A (or about 3 kyr of stratigraphic time). In the remainder of the zone, the characteristic directions are remarkably antiparallel.

Closer look at the evolution of the susceptibility as a function of heating reveals some features which are at variance with those observed outside of the complex zones (Fig. 6). Whereas in the latter susceptibility generally remains constant up to 600°C, or in any case never exceeds room-temperature susceptibility by more than 50% after heating to 550°C, all samples from zones A and B display susceptibility increases beyond 400°C, reaching at 500°C up to 3 times the initial susceptibility. This indicates that some mineralogical change has occurred beyond 400°C which may be due to a change in the type of magnetite or titanomagnetite (a contribution of iron sulfides is not consistent with unblocking temperatures higher than 430°C). In any case, this means that magnetic directions above that temperature should be regarded with caution, despite the fact that in a well shielded furnace they might remain meaningful.

Indeed inspection of vector demagnetization diagrams reveals two very different types of behaviour. In some samples (Fig. 7), a clear linear component decaying almost to the origin is isolated (after removal of a PEF overprint) between 300 and 450°C. The linear plots miss the origin by less than 10% of the remaining magnetization at 300°C. The temperature range of the characteristic component is similar to that of the components isolated outside the complex zones.

Other samples show a very distinct demagnetization (Fig. 8). A clear component is removed between 120 and 500–550°C (Fig. 8, a and b) or between 15 and 60 mT (Fig. 8c), i.e. in the same range as seen above (Fig. 7). However, significantly more than half of the NRM remaining at 120°C still persists at 500–550°C. Moreover, there is an indication that in some samples, a component with normal polarity is removed before 450°C or 60 mT, before the removal of a component with reverse polarity between 550 and 600°C (Fig. 8, a and b) or between 70 and 90 mT (Fig. 8c). It seems quite reasonable to assume that this HT, high coercivity component, which is absent in the rest of the section, is an abnormal magnetization, with a very narrow spectrum of grain characteristics.

A third kind of observation lies with the magnetic intensity remaining after heating at 300°C but prior to any conspicuous mineralogical change (Fig. 5). This is found to be scattered by one order of magnitude or less at any given level and to increase monotonously on average from the bottom to the top of the section (by one order of magnitude), particularly in the zone from −70 to −20 m. The outstanding exception is zone A, where the scatter in intensity reaches two orders of magnitude and where the highest values are reached (the other strong magnetizations are found 35 m above in the pink and red marlstones, immediately under and above the KTB). It is interesting to note that the distribution of intensities is different for the two polarities: 11 reversed samples, as opposed to only 3 normal samples have magnetization larger than 5.10^{-4} A/m, whereas 30 normal samples, as opposed to 7 reversed ones have magnetization lower than 10^{-4} A/m. Thus, the stronger samples tend to be the reversed ones. The two samples shown in Fig. 7 are at 1 to 3.10^{-4} A/m (after a thermal cleaning up to 300°C); three out of four samples in Fig. 8 are between 4 and 10.10^{-4} A/m.

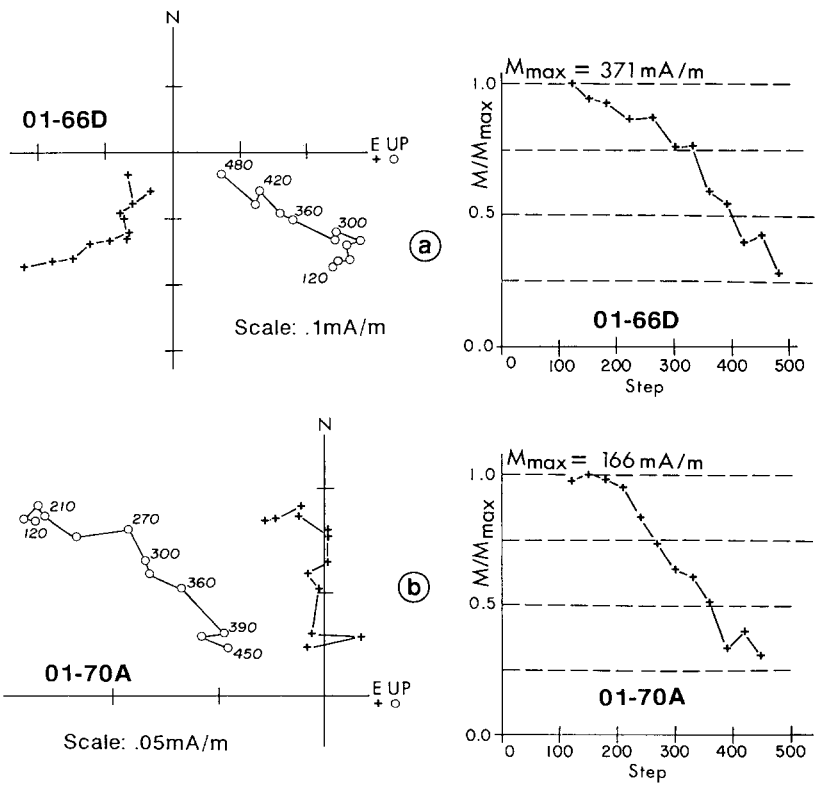


Fig. 7. Representative vector demagnetization diagrams and NRM decay curves for (a) reversed and (b) normal samples from complex zone A (stratigraphic coordinates). Same symbols as in Fig. 3.

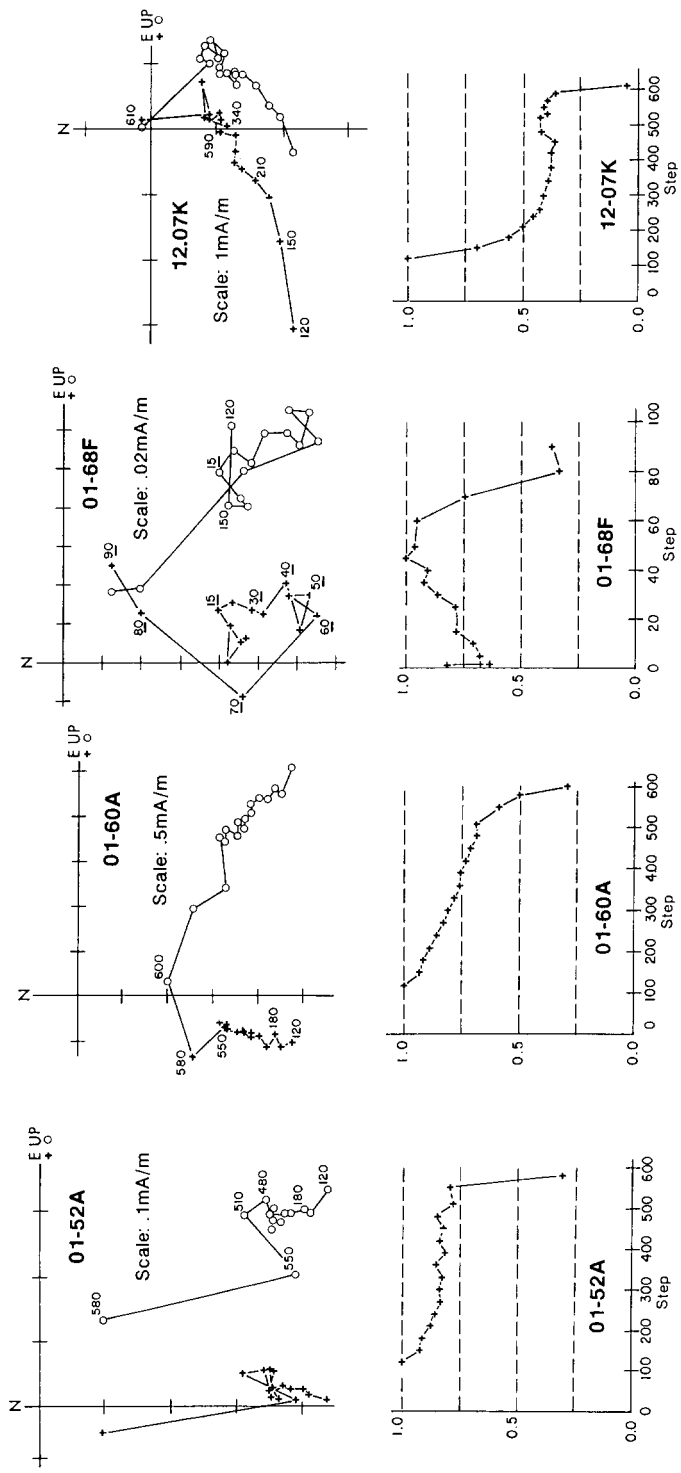


Fig. 8. Further representative vector demagnetization diagrams and NRM decay curves for the complex zones (stratigraphic coordinates). (a) Sample with HT component. (b) Sample with two parallel components. (c) Sample with two antiparallel components. (d) Sample from zone B, with PEF and HT components. Same symbols as in Fig. 3.

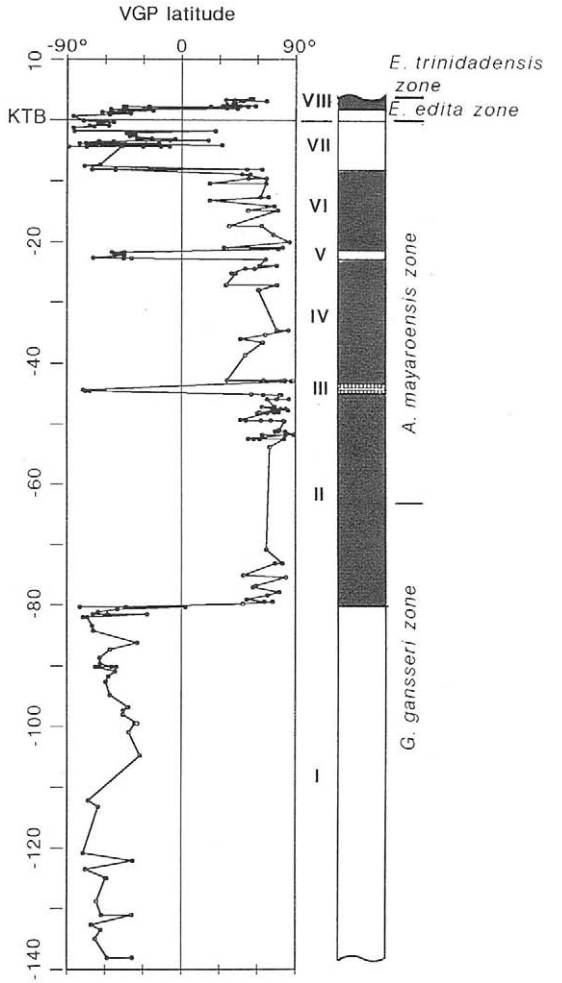


Fig. 9. Proposed magnetostratigraphy for the Sopelana section. Variations of VGP latitude (after tectonic correction) and biostratigraphic zonation are given. Polarity zones are numbered from I to VIII.

We have decided to discard all directions corresponding to a combination of high intensities, existence of a HT component and/or increase of susceptibility above 450°C, as being most likely remagnetized. We have retained only samples that displayed a clear lower temperature component in the range characteristic of the rest of the section, with demagnetization being achieved by 450°C. As a result, all the samples within complex zone B were discarded. Indeed, none of them was demagnetized by 500°C and no lower temperature component different from the PEF could be isolated (e.g. Fig. 8d). Fourty-three out of 110 samples were retained in complex zone A (Fig. 9), most of them with normal polarity, except between −43 and −45 m. In these 2 m, only three samples (from the same stratigraphic level) become demagnetized before 450°C, and these three samples have a clear reversed polarity (Fig. 7a). The thickness of this LT reversed zone is rather uncertain, ranging between 0.2 and 2 m. We suggest that this zone displays the altered remains of chron 30R, subsequently partly (but not in all samples) overprinted during chron 30N time. It is indeed reasonable to assume that it is chron 30R that is responsible for the intense, complex, largely reversed signatures in a 10 m interval of the section (all these HT components with reverse polarity have been found in samples taken between −49 m and −43 m). In the same way, we can assume that the HT magnetization of the zone B samples has been overprinted during the short reverse polarity event found at −22 m below the KTB. An X-ray diffraction study of the sediments from the Sopelana section (between −60 and −20 m below the KTB) shows a much larger amount of chlorite in zones A and B than elsewhere (I. Cojan, pers. commun., 1990). This may indicate more intense diagenesis in these two zones than elsewhere in the section, with possible remagnetization of zone A in a later reversed interval such as 30R or 29R.

5. Discussion and conclusion

Our final selection of 223 reliable characteristic directions over the 150 m of strata studied at Sopelana is shown in Fig. 9. It outlines a succession of eight polarity intervals (numbered in roman numerals from bottom (I) to top (VIII). The *A. mayaroensis* zone, which characterizes the upper Maastrichtian, is entirely represented within the section. Its FAD lies within magnetic zone II, which must therefore correspond to chron 31N; its LAD is at the KTB, in zone VII, which is identified as chron 29R [20, 26]. The three samples with positive VGP latitudes within 29R correspond to within-strata shear in two cases and an isolated, unreproducible result in one case, as discussed above. The reversed zone I in the *G. gansseri* biostratigraphic zone (with *G. stuarti* present) must correspond to the end of reversed chron 31R and the normal zone VII must be chron 29N. It is interesting to point out that *E. trinidadensis* appears in that chron [28, 29]. We are left, as discussed in the body of the paper, to identify chrons

31N, 30R and 30N within magnetic zones II to VI. We observe that there is one extra short reversed zone compared to the generally accepted reversal succession (Fig. 10). There is no biostratigraphic argument that constrains our choice of a correlation of 30R with either III or V. We have noted that III was likely to be over twice as thick (as long) as V. Moreover, a comparison of our section with the reference section at Gubbio (Fig. 10, b and c; [1]), where the FAD of *A. mayaroensis* and the KTB have been aligned for comparison (and normalization) shows that III correlates well with 30R at Gubbio. Assuming that sedimentation rates are approximately uniform or undergo similar fluctuations at both places, we propose to identify III (remembering its complex remagnetization effects) with 30R. We realize however that hypotheses on sedimentation rate are always hazardous until documented independently.

As a result, we are led to propose the existence of a reversed event, V, with a duration probably less than 30 kyr (1 m thickness), within chron 30N. Neither this short event, nor the 30R chron have been identified in the Moria section [2],

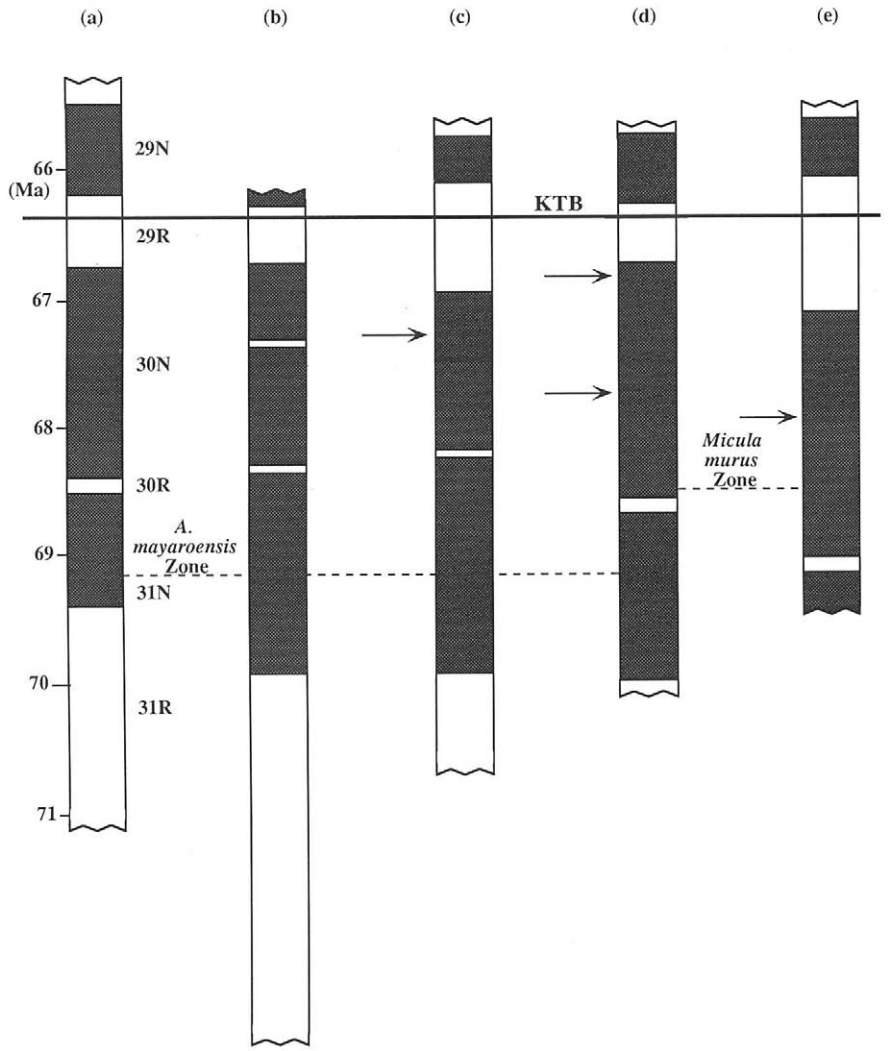


Fig. 10. Correlation between magnetostratigraphic columns and the magnetic reversal time scale. (a) Berggren scale [20, 21], (b) Sopelana section (this paper, see Figs. 5 and 9), (c) Gubbio section [30,31], (d) DSDP Hole 524 [4,5], (e) DSDP Hole 577A [3]. (b), (c) and (d) are aligned at the KTB (solid line) and at the FAD of *A. mayaroensis* (lower dashed line); (d) and (e) are aligned at the KTB and at the FAD of *M. murus*. Arrows: unidentified short magnetic events.

where the sedimentation rate is half that at Sopelana. On the other hand, in the Gubbio section and despite a sedimentation rate four times less than at Sopelana, the most complete accounts [30, 31] reveal one data point with a negative VGP latitude in the upper part of chron 30N (Fig. 10c).

Moreover, DSDP Site 524 [4, 5], with a sedimentation rate similar to that at Sopelana, shows at least one reversed sample at a level that could correspond to event V (Fig. 10d; sampling rate in that core is about once per 10 kyr). At DSDP Hole 577A [3], where sedimentation rate is similar to that at Gubbio, reversed samples are also observed at the appropriate level within 30N (Fig. 10e). We therefore find some independent support for the existence of a short reversed event approximately halfway up chron 30N. This would provide an interesting new tie-level within this key period of time and also imply a somewhat higher reversal frequency [32]. An alternate interpretation would be to discard all samples from complex zone A as likely to have been remagnetized, and to interpret zone V as the recording of chron 30R. This, which is not our favoured solution for reasons developed above, would imply exceedingly non-uniform sedimentation rates.

In conclusion, the very high sedimentation rate at Sopelana, coupled with dense paleomagnetic sampling, provides a reasonable magnetostratigraphy which is well tied to the biostratigraphy. Hence, this should be a useful addition to the still limited collection of detailed magnetostratigraphies in the neighbourhood of the KTB. The 29N to 31R sequence has a spatial (hence temporal) resolution which is unmatched by other subaerial sections and is only found in DSDP Site 524. It is the first successful magnetostratigraphy with that time range and resolution within the exceptionally thick and well-studied KTB sequence of the Basco–Cantabric basin. Its resolution allows us to propose the existence of an additional short reversed event within chron 30N, but also to discard (at the resolution level) the presence of other events between the top of 29R and the middle of 31R. Yet this magnetostratigraphy did not come without problems. Although, thanks to wide zones with clear magnetic signal and good biostratigraphic control, there is no doubt in the identification of most magnetochrons, we found two exceedingly complex zones of remagnetization, where apparently good reversed and normal directions were found in impossibly quick succession. Although magnetic analysis did reveal some differences in local magnetic behaviour and could hint towards remagnetization, this could well have escaped attention if sampling had been coarser (or sedimentation rate much lower), and if good biostratigraphic check had not been available. It might be useful to check earlier published magnetostratigraphies where such phenomena might have escaped attention. We see in Fig. 5 that the lower remagnetized part of complex zone A and the entire complex zone B could have been overprinted respectively at the time of deposition of short reversed magnetic zones III and V, of which they would be a sort of "echo". These 1 to 3 m thick zones occur approximately 3 m below the remagnetizing chron. This means that about 100 kyr would have elapsed with the sediments, still rich in water, many metres below the water–sediment interface when remagnetization took place. The causes of this remagnetization are however still elusive. A more complete coupled magnetic, mineralogical and sedimentological study is in progress.

Acknowledgements

We are grateful to W. Lowrie and three anonymous reviewers for useful comments. This is contribution IPGP NS 1175.

References

1 W. Alvarez, M.A. Arthur, A.G. Fisher, W. Lowrie, G. Napoleone, I. Premoli Silva and W.M. Roggenthen, Upper Cretaceous Paleocene magnetic stratigraphy at Gubbio, Italy, Geol. Soc. Am. Bull. 88, 367–389, 1977.

2 W. Alvarez and W. Lowrie, Upper Cretaceous paleomagnetic stratigraphy at Moria (Umbrian Appennines, Italy): verification of the Gubbio section, Geophys. J.R. Astron. Soc. 55, 1, 17, 1978.

3 U. Bleil, The magnetostratigraphy of Northwest sediments, Init. Rept. DSDP LEG 86, 441–458, 1984.

4 R.Z. Poore, L. Tauxe, S.F. Percival, J.L. LaBrecque, R. Wright, N.P. Petersen, C.C. Smith, P. Tucker and K.J. Hsü, Cretaceous–Cenozoic magnetostratigraphic and biostratigraphic correlations of South Atlantic Ocean: DSDP 73, Palaeogeogr., Palaeoclimatol., Palaeoecol. 42, 127–149, 1983.

5 L. Tauxe, P. Tucker, N.P. Petersen and J.L. LaBrecque, The magnetostratigraphy of Leg 73 sediments, Palaeogeogr., Palaeoclimatol., Palaeoecol. 42, 65–90, 1983.

6 S.V. Margolis, J.F. Mount, E. Doehne, W. Showers and P. Ward, The Cretaceous/Tertiary boundary, carbon and oxygen isotope stratigraphy, diagenesis and paleoceanography at Zumaya, Spain, Paleoceanography 2, 361–377, 1987.
7 J.F. Mount, S.V., Margolis, W. Showers, P. Ward and E. Doehne, Carbon and oxygen isotope stratigraphy of the upper Maestrichtian, Zumaya, Spain: a record of oceanographical biologic changes at the end of the Cretaceous period, Palaios 1, 87, 1986.
8 J.F. Mount and P. Ward, Origin of limestone/marl alternations in the Upper Maastrichtian of Zumaya, Spain, J. Sediment. Petrol. 56, 228–236, 1986.
9 S.F. Percival and A.G. Fischer, Changes in calcareous nannoplankton in the Cretaceous–Tertiary biotic crisis at Zumaya, Spain, Evol. Theory 2, 1–35, 1977.
10 W.M. Roggenthen, Magnetic stratigraphy of the Paleocene. A comparison between Spain and Italy, Mem. Soc. Geol. Ital. 15, 73–82, 1976.
11 P. Ward, J.Weidman and J.F. Mount, Maastrichtian molluscan biostratigraphy and extinction patterns in a Cretaceous/Tertiary boundary section exposed at Zumaya, Spain, Geology 14, 899–903, 1986.
12 R. Rocchia, D. Boclet, P. Bonté, E. Buffetaut, X. Orue-Etxebarria, J.J. Jaeger and C. Jéhanno, Structure de l'anomalie de l'iridium à la limite Crétacé–Tertiaire du site de Sopelana (Pays Basque espagnol), C.R. Acad. Sci. Paris 307, 1217–1223, 1988.
13 Ph. Bonté, O. Delacotte, M. Renard, C. Laj, D. Boclet, C. Jehanno and R. Rocchia, An iridium rich layer at the Cretaceous–Tertiary boundary in the Bidart section (Southern France), Geophys. Res. Lett. 11, 473–475, 1984.
14 O. Delacotte, M. Renard, C. Laj, K. Perch-Nielsen, I. Premoli-Silva and S. Clauser, Magnétostratigraphie et biostratigraphie du passage Crétacé–Tertiaire de la coupe de Bidart (Pyrénées-Atlantiques). Bull. B.R.G.M., 1, Géol. de la France 3, 243–254, 1985.
15 M.A. Lamolda, X. Orue-Etxebarria and F. Proto-Decima. The Cretaceous–Tertiary boundary in Sopelana (Biscay, Basque Country), Zitteliana 10, 663–670,1983.
16 X. Orue-Etxebarria, Los Foraminiferos planctonicos del Paleogeno del Sinclinorio de Bizkaia (Corte de Sopelana–Punta de la Galea), Parte 1: Kobie (Bilbao) 13,175–248, 1983; Parte 2: Kobie (Bilbao) 14, 351–429, 1984.
17 J.C. Plaziat, Signification paléogéographique des "calcaires conglomérés" des brèches et des niveaux à Rhodophycées dans la sédimentation carbonatée du bassin Basco–Béarnais à la base du Tertiaire (Espagne–France). Rev. Géogr. Phys. Géol. Dyn. 17, 239–258, 1975.
18 P. Rat, Les Pays Crétacés Basco-Cantabriques, 125 pp., Thèse, Fac. Sci. Dijon, Publ. Univ. Dijon 18, 1959.
19 P. Ward, Maastrichtian ammonite and inoceramid ranges from Bay of Biscay. Cretaceous–Tertiary boundary sections. Rev. Espan. Paleontol., N° Extraordinario, 119–126, 1988.
20 W.A. Berggren, D.V. Kent, J.J. Flynn and J.A. Van Couvering, Cenozoic geochronology, Geol. Soc. Am. Bull. 96, 1407–1418, 1985.
21 D.V. Kent and F.M. Gradstein, A Cretaceous and Jurassic geochronology, Geol. Soc. Am. Bull. 96, 1419–1427, 1985.
22 V.D. Herm, A. Hillebrandt and K. Perch-Nielsen, Die Kreide/Tertiär-Grenze im Lattengebirge (Nördliche Kalkalpen) in mikropalaeontologischer Sicht, Geol. Bavarica 82, 319–344, 1981.
23 G. Keller, Extinction, survivorship and evolution of planktic foraminifera across the Cretaceous/Tertiairy boundary at El Kef, Tunisia, Mar. Micropaleontol., 13, 239–263, 1988.
24 J.L. Kirschvink, The least square line and plane and the analysis of paleomagnetic data, Geophys. J.R. Astron. Soc. 62, 699–718, 1980.
25 J. Besse and V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma, J. Geophys. Res., in press.
26 W.B. Harland, R.L. Armstrong, A.V. Cox, L.E. Craig, A.G. Smith and D.G. Smith (eds.), The magnetostratigraphic time scale, in: A Geologic Time Scale, Cambridge University Press, 140–167, 1989.
27 T.D. Herbert and S.L. D'Hondt, Precessional climate cyclicity in Late Cretaceous–Early Tertiary marine sediments: a high resolution chronometer of Cretaceous–Tertiary boundary events, Earth Planet. Sci. Lett. 99, 263–275, 1990.
28 J.J. Groot, R.B.G. de Jonge, C.G. Langereis, W.G.H.Z. ten Kate and J. Smit, Magnetostratigraphy of the Cretaceous–Tertiary boundary at Agost (Spain), Earth Planet. Sci. Lett. 94, 385–397, 1989.
29 W.A. Berggren, Comments on the paper "Magnetostratigraphy of the Cretaceous–Tertiary boundary at Agost (Spain)", J.J. Groot, A.B.G. de Jonge, C.G. Langereis, W.G.H.Z. ten Kate and J. Smit, Earth Planet. Sci. Lett. 95, 183–185, 1989.
30 W. Lowrie and J.E.T. Channell, A review of magnetic stratigraphy investigations in Cretaceous Carbonate rocks, J. Geophys. Res. 85, 3597–3605, 1980.
31 W. Lowrie and F. Heller, Magnetic properties of marine limestones, Rev. Geophys. Space Phys. 20, 171–192, 1982.
32 W. Lowrie and D.V. Kent, Geomagnetic reversal frequency since the Late Cretaceous, Earth Planet. Sci. Lett. 62, 305–313, 1983.